

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think> For part (a), we need to calculate the heat required to raise the temperature of 1 liter of water (which is 1000 grams) from 30°C to 60°C . The formula for heat is $Q = m * c * \Delta T$, where m is mass , c is specific heat , and ΔT is the change in temperature . For water , m = 1000 g , c = 1 cal/g °C , and $\Delta T = 60^{\circ}\text{C} - 30^{\circ}\text{C} = 30^{\circ}\text{C}$. So , $Q = 1000 * 1 * 30 = 30000 \text{ cal} = 30 \text{ Kcal}$.

For part (b), we need to calculate the heat required to raise the temperature of 1 kg of aluminum (which is 1000 grams) from 30°C to 60°C . For aluminum , m = 1000 g , c = 0.215 cal/g °C , and $\Delta T = 30^{\circ}\text{C}$. So , $Q = 1000 * 0.215 * 30 = 6450 \text{ cal} = 6.45 \text{ Kcal}$.

Therefore , the answer is 30 Kcal , 6.45 Kcal .

</think><answer> 30 Kcal , 6.45 Kcal </answer>

